# Venkata Lakshmi Parimala Pasupuleti

Backend Engineer · Python · Document Understanding · RAG & Ingestion Pipelines · K8s · Temporal · Terraform · LLM-Native Applications

venkatalakshmiparimala@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

## PROFESSIONAL SUMMARY

Backend engineer with 8+ years building production systems at scale, and the last two years building LLM-native applications in the exact problem space LlamaIndex owns. Leasel is a document ingestion and understanding platform: OCR-fallback ingest, section-aware chunking, hybrid search (TF-IDF + semantic reranking), citation grounding to exact source clauses, and a full evaluation harness — the same building blocks LlamaIndex ships as infrastructure. AgentProbe is an open-source AI eval toolkit built from scratch because the tooling didn't exist. Both went from prototype to production. Backend foundations: Python (8+ yrs), PostgreSQL, K8s, Terraform, Temporal, RabbitMQ, AWS — owned 20+ modules serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

## WORK EXPERIENCE

**Blue Horse Digital** Sep 2026 – Present

**AI/ML Engineer (Volunteer) · Remote**

- Building ML models and data APIs; integrating AI components across backend and reporting layers; debugging cross-stack reliability issues in data-driven features.

**Magizhs Tech Solution** May 2025 – Aug 2025

**AI/ML Engineer Intern · Remote**

- Built an agentic LangGraph pipeline (parse → plan → retrieve → validate → conditional route) with RAG over FAA Part 107 corpus — natural-language requests into structured outputs with cited-source compliance answers; guardrails blocking non-compliant missions before export.

**George Mason University** Aug 2024 – Aug 2025

**Research Associate + Teaching Assistant**

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries ($R^2$=0.977); HuggingFace fine-tuning + inference API deployment; Isolation Forest anomaly detection; statistical validation. Workshops to 600+ students across Python and SQL.

**Entrata (PropTech / Real Estate SaaS)** Sep 2018 – Aug 2024

**Senior Software Engineer · Remote**

- **Backend owner across 20+ production modules end-to-end** (Python, PHP, Java, PostgreSQL, RabbitMQ, AWS S3/EC2/RDS) serving 100K+ monthly active users at 99.5% uptime — designed, built, tested, deployed, operated, and fixed when things broke. 120+ Agile sprints. Owned a mission-critical compliance pipeline solo for 7 years: no spec, no PM, no playbook. Zero violations. Zero security incidents.
- **Distributed systems built for correctness first** — event-driven architecture (RabbitMQ) with retry logic and idempotency guarantees; multi-tenant PostgreSQL for 100K+ housing units; 200+ SQL optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement, 40% deployment time reduction). Temporal-based async processing for critical state workflows.
- **Infrastructure as code and observability** — Terraform IaC; Kubernetes deployments; OpenTelemetry + Prometheus + Grafana (40% reduction in debugging time); AWS Lambda + Step Functions for orchestration; Python/Bash automation eliminating 80% of manual operational workload.

**Entrata India, Pune** Jul 2017 – Sep 2018

**Associate Software Engineer**

- Built property management backend modules (PHP, Python, Java, PostgreSQL); led AWS S3 migration; data pipelines across hundreds of thousands of records.

## PROJECTS

**Leasel: Document Understanding & Ingestion Platform** RAG + INGESTION LANGGRAPH

Python · LangGraph · FastAPI · TF-IDF · OCR · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- End-to-end document understanding and AI investigation platform sharing one evidence model: OCR-fallback ingestion, section-aware chunking, **hybrid search** (TF-IDF + semantic reranking), citation grounding to exact source clauses, missing/conflicting clause detection, and human-review gate on regulated decisions. 8-agent LangGraph pipeline with durable state, conditional routing, and fault-tolerant HITL escalation. Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 lease PDFs (5 US states + federal); pass@k harness, LLM-as-judge scoring, CI regression gate. LLM reasons; deterministic logic decides every regulated outcome.

**AgentProbe: Open-Source AI Evaluation Infrastructure** CI-INTEGRATED OPEN SOURCE

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Node.js | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine; LLM-as-judge scoring; CI regression harness with golden datasets; distributed model routing across OpenAI/Anthropic/Groq/Ollama; offline + online evaluation loops. Built from scratch because the tooling didn't exist — the same instinct LlamaIndex describes: identify what the ecosystem needs, build it, ship it.

## EDUCATION

**M.S. Data Analytics Engineering** George Mason University, Virginia May 2026

GPA: 3.87

**B.Tech Computer Science & Engineering** JNTU Anantapur 2017

## SKILLS

**Languages & Frameworks:** Python (8+ yrs), JavaScript, TypeScript, Node.js, Java (Spring Boot), PHP, SQL, PL/SQL, Bash — FastAPI, Django, React.js, GraphQL, REST APIs

**Backend & Infrastructure:** PostgreSQL, Redis, RabbitMQ, Apache Kafka, Temporal, Microservices, Event-Driven Architecture, Idempotency, Fault-Tolerant Design, Docker, Kubernetes (K8s), Terraform, AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), GitHub Actions CI/CD

**AI & Data:** LangGraph, LangChain, RAG Pipelines, Document Understanding, Ingestion Pipelines, Hybrid Search, LLM Evaluation, AgentProbe, Agentic Workflows, OCR, Semantic Chunking, Embeddings, ChromaDB, OpenTelemetry, Prometheus, Grafana, PyTorch, HuggingFace